

Matteo He

Research interests: *AI Safety, Alignment, and Interpretability*
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Research Profile

Early-career researcher interested in the safety and alignment of advanced AI, including detecting hidden objectives, evaluating behavior under oversight, and mechanistic interpretability. Cambridge MPhil with Distinction and AI Alignment Fellowship. Dissertation work led to a first-author arXiv preprint and a related manuscript under review on language-model readouts, with public code and pretrained artifacts. Previously built ML-evaluation infrastructure at Amazon.

Selected Research

The two readout projects below arose from my Cambridge MPhil dissertation.

Sparse Readout Prism [arXiv:2609.01936] [code] [models] First-author preprint; under review

- Developed SRP to decompose selected logits into sparse feature contributions and explicit residuals. Across six readouts without softcaps, improved logit-difference coverage by **8.9–17.3 percentage points** over the strongest of six geometric baselines; coverage requires the correct sign and less than 50% relative error.
- Showed that fitted lenses could change their reported surface language while the dominant SRP feature remained stable in **77/80 prompts (96%)**; reproduced the result with English-German and ridge-lens controls.

Learning to Read Out: Unembedding Dynamics [code] [artifacts] First-author submission; under review

- Tracked how sparse output features form, reorganize, and become load-bearing across public pretraining checkpoints using trajectory crosscoders; separated hidden-state availability from native-readout expression with readout swaps, controlled probes, and ablations.
- Released a tested checkpoint-analysis library with experiment-to-figure manifests, guided notebooks, reproducible configs, and pretrained parameter-trajectory crosscoders.

Automated Interpretability of GPT-2 Attention Heads BSc dissertation, 2024

- Built a PyTorch/Transformers explainer-simulator pipeline; improved distribution-based evaluation metrics by **25%**, cut token errors from **366 to 157 (57%)**, and generated high-fidelity explanations across **10+** syntactic attention heads.

Education & Distinctions

University of Cambridge	Cambridge, UK
<i>MPhil in Advanced Computer Science, Distinction; AI Alignment Fellowship</i>	<i>Oct 2025 - Jun 2026</i>

University of St Andrews	St Andrews, UK
<i>BSc (Hons) Computer Science and Mathematics, First Class; GPA 18/20 (90%); Top Student Medal</i>	<i>Sep 2021 - Jun 2024</i>

GRE: 340/340 (Quantitative: 170/170; Verbal: 170/170). Dean's List in all three years at St Andrews.

Selected Research & ML Experience

Japanese Language Learning Platform	Jun 2024 - Oct 2025
<i>Founder & ML/Software Engineer (platform now archived)</i>	<i>Remote</i>

- Solo-built a privacy-preserving browser LLM platform serving **about 3K MAU**; moved inference on-device with WebGPU/WebLLM, cut p95 latency by **40%**, and maintained **approximately 99.9% uptime**.

Amazon, Alexa-AI (NLU)	Jun 2023 - Sep 2023
<i>Software Development Engineer Intern (ML Evaluation Infrastructure)</i>	<i>London / Cambridge, UK</i>

- Built Python/AWS evaluation infrastructure for **20+ Applied Scientists**, processing **millions of utterances daily**; created failure-point analysis for model debugging, accelerating iteration by **15%** and reducing manual validation by **40%**.

St Andrews Research Internships	2022 - 2023
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- Improved dolphin-acoustics classification F1 from **0.48 to 0.86** and refactored a **5K+ line** Python research codebase; automated collection of **600+ hours** of solver metrics for energy-aware algorithm selection.

Technical Methods

Research: mechanistic interpretability, readout auditing, sparse coding, representation analysis, LLM evaluation, readout swaps, ablations, reproducibility

Technical: Python, PyTorch, Transformers, TransformerLens, JAX, GPU training, AWS, distributed experiment pipelines

Languages: English, Italian, and Mandarin Chinese (native/bilingual); French and Spanish (professional)